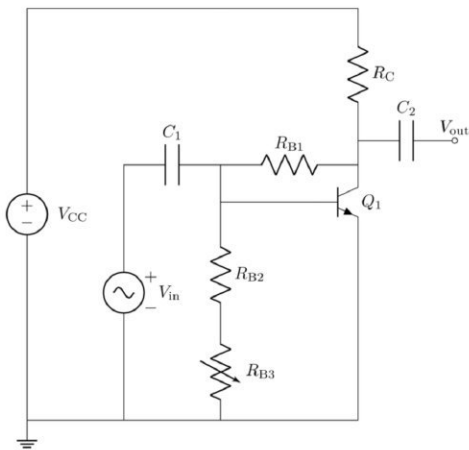


Section 3: Optimisation of a different configuration of CE amplifier

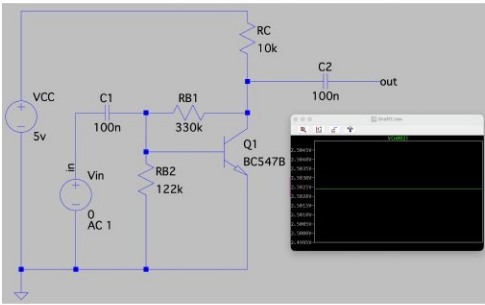
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Circuit used:

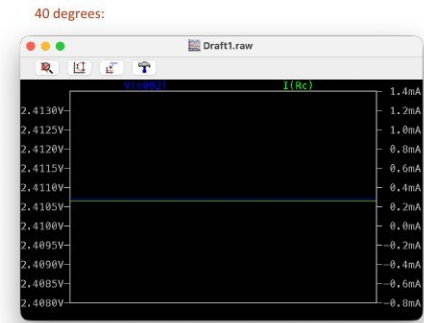


Operating Point

Use LTspice to build the circuit, set values of RC, RB1, RB2, and RB3 so that the output = 2.5V



Use LTspice to find a suitable biasing for the amplifier. Find the temperature stability of the bias point and the power consumption.



	A	B	C	D
1	Temperature/degree	Vout/V	Supply I/mA	Power/mW
2	0	2.684	0.227	1.135
3	5	2.65	0.228	1.14
4	10	2.616	0.232	1.16
5	15	2.582	0.238	1.19
6	20	2.548	0.239	1.195
7	25	2.514	0.243	1.215
8	30	2.479	0.248	1.24
9	40	2.411	0.255	1.275
10	50	2.342	0.258	1.29
11	60	2.273	0.268	1.34
12	70	2.204	0.274	1.37
13	80	2.134	0.281	1.405
14	85	2.1	0.283	1.415

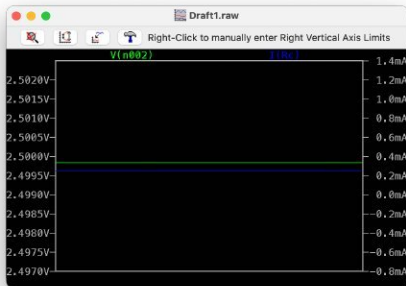
Gain

You can measure the gain of an amplifier in simulation by running an AC analysis. That will show

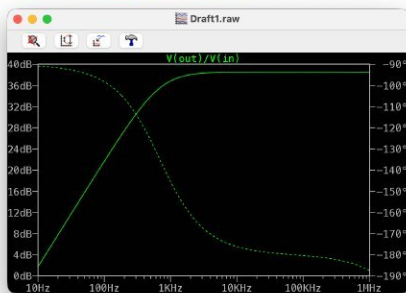
you the amplitude of signals as a function of frequency, including the transfer function. Use the 'Add trace' option to add a plot of V_{out}/V_{in} to the graph. You will see that the gain is constant with frequency, except for a roll-off at low frequency due to the high-pass filter made from C_1 and the input impedance of the amplifier.

Find the gain of the amplifier. Try changing the bias resistors to see how they affect gain. Optimise the circuit to achieve a good trade-off between gain and power consumption, whilst maintaining a quiescent voltage of 2.5V at V_C . Investigate how gain is affected by temperature.

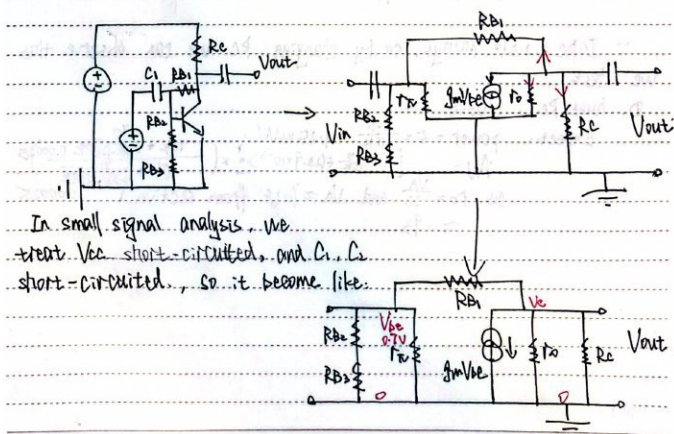
Check for 2.5V bias:



In the transfer function we can see, gain is stable at about 39dB with frequency changes. Initially, gain increases from 0 because of the capacitor.



Trade-off:



In small signal analysis, we treat V_{cc} short-circuited, and C_1, C_2 short-circuited, so it become like:

Take KCL at collector:

$$\frac{V_c}{R_c} + \frac{V_c}{R_L} + \frac{V_c - V_{be}}{R_{B1}} + g_m V_{be} = 0$$

$$\therefore V_c = \left(\frac{V_{be}}{R_{B1}} - g_m V_{be} \right) \div \left(\frac{1}{R_c} + \frac{1}{R_L} + \frac{1}{R_{B1}} \right)$$

$$\therefore \text{Gain} = A_v = \frac{V_{out}}{V_{in}} = \frac{V_c}{V_{be}} = \frac{\left(\frac{V_{be}}{R_{B1}} - g_m V_{be} \right)}{\left(\frac{1}{R_c} + \frac{1}{R_L} + \frac{1}{R_{B1}} \right)} \div V_{be}$$

$$= \frac{1 - g_m R_{B1}}{\frac{R_{B1}}{R_c} + \frac{R_{B1}}{R_L} + 1} \quad \text{where } g_m = \frac{I_c}{V_T}$$

$$\approx \frac{-g_m R_{B1}}{\frac{R_{B1}}{R_c} + \frac{R_{B1}}{R_L}}$$

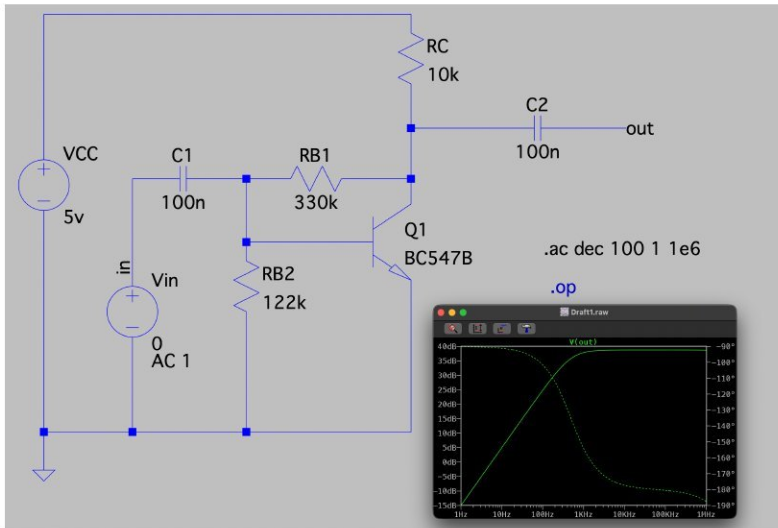
$$\approx -g_m (R_c \parallel R_L) \quad \text{which is consistent as book}$$

$$\text{So when } I_c \uparrow, A_v \uparrow \approx -\frac{I_c}{V_T} (R_c \parallel R_L)$$

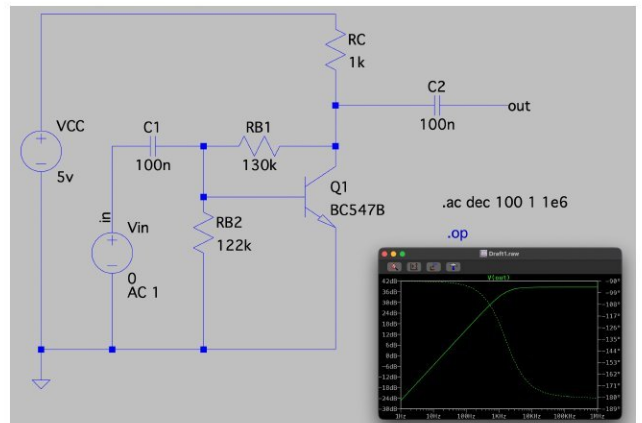
$$\text{The other side: Power consumption} = I V$$

$$= I_c V_{cc}$$

So when $I_c \uparrow$, consumption $\uparrow$.
 There is a trade-off.
 $\therefore I_{CQ} = 2.51\text{mA}$ always... so by changing R_C we can observe this we test.
 D. huge $R_C = 10\text{k}\Omega$.
 Expect: power = $5 \times \frac{2.5}{10\text{k}} = 1.25\text{mW}$.
 $A_v = \left[-\frac{10\text{k}}{10\text{k} + (2.5 \times 10^{-3})} \right] \times \left(\frac{10\text{k} \times \frac{10}{10\text{k} + \frac{10}{100\text{n}}}}{10\text{k} + \frac{10}{100\text{n}}} \right)$
 as $r_o = \frac{V_A}{I_{CQ}}$ and $V_A \approx 10\text{V}$ from section 1.
 ≈ -93 .

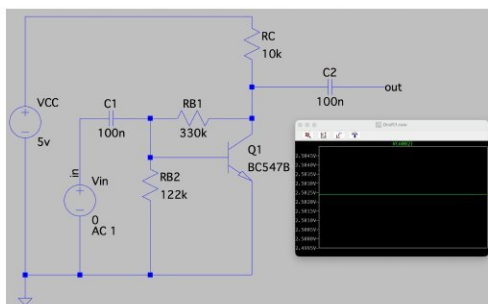


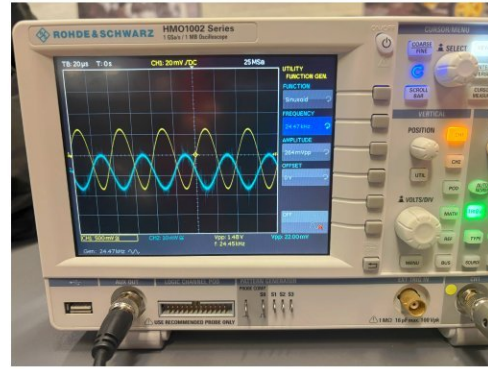
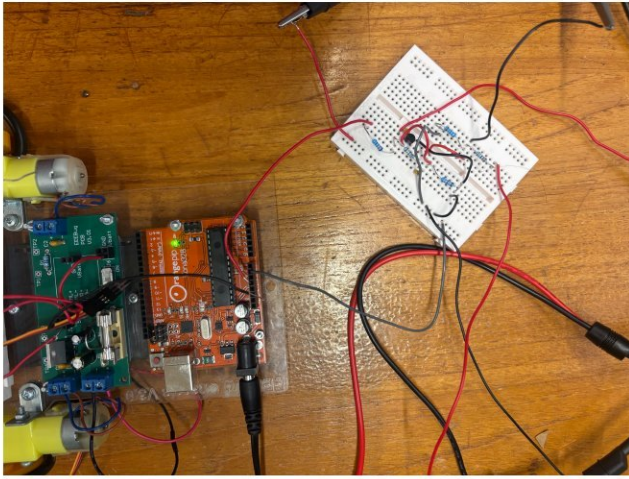
Small $R_C = 1\text{k}\Omega$.
 Power = 6.25mW .
 $A_v = -93.2 \Rightarrow$ same calculation.
 $\uparrow 1.25\text{mW} \rightarrow \uparrow 6.25\text{mW} \uparrow$
 $\uparrow -93 \rightarrow \uparrow -93.2 \uparrow$ pattern is correct.
 $\Rightarrow 39.3\text{dB} \Rightarrow 39.3\text{dB}$.
 In LTspice.
 $R_C = 10\text{k}\Omega$: 38.7dB & $1.225 \times 10^{-3}\text{mW}$.
 $R_C = 1\text{k}\Omega$: 38.9dB & $12.51 \times 10^{-3}\text{mW}$.
 which is correct and close in trend.
 One interesting finding is that although gain doesn't change much, waste has a large difference.
 So we choose a large $R_C = 10\text{k}\Omega$ to make amp.



Challenge: Build Your Amplifier

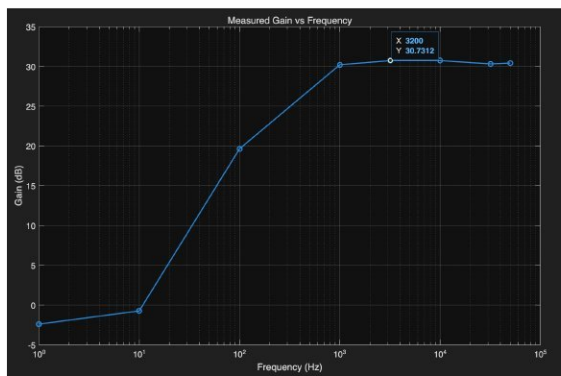
Once you are happy with the component values in your amplifier, build it on your breadboard. You will need to find the closest resistor values available in the lab. Check the bias conditions and measure the transfer function to compare your circuit with the simulation.





From the circuit we got data and transfer function like:

A	B	C	D
Frequency/Hz	Vin/mV	Vout/mV	Gain/dB
1	10	7.6	-2.383728154
10	10	9.2	-0.724243453
100	10	96	19.64542466
1000	10	324	30.2109002
3200	10	344	30.73116885
10000	10	344	30.73116885
32000	10	328	30.31747687
50000	10	332	30.42276167



Although gain is 30.7dB, in Ltspice it is about 39dB.

It is acceptable.

Some possible uncertainty due to:

- We used OrangePi to power BJT, which is not very stable and contains transistors.
- Internal resistance of capacitors & signal generators. Etc.